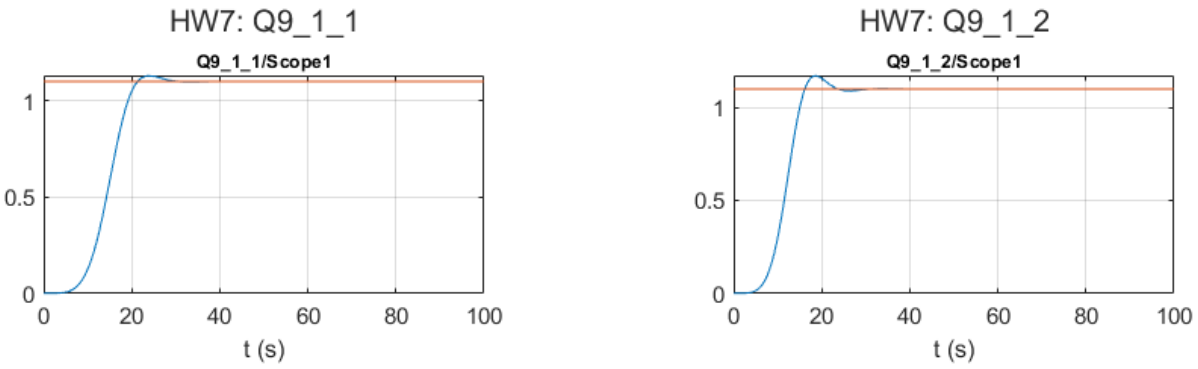
Gain Scheduling: Compensating a Nonlinear Valve (Åström & Wittenmark, Example 9.1)

Adaptive Control, Homework 7

Arshia Goshtasbi • MSc Electrical Engineering (Control), Iran University of Science and Technology • Autumn 1404 (2025)

Both cases are subsystems of `HW7.slx`. A process is driven through a valve with a nonlinear characteristic. Following the book, I compensated the valve with an approximate inverse of its characteristic, built as a piecewise-linear function with four segments, so that the loop sees a roughly linear actuator. I simulated two versions of the loop and plotted the output against the reference.



Re-run in R2023b: the two sub-models of `HW7.slx`.